

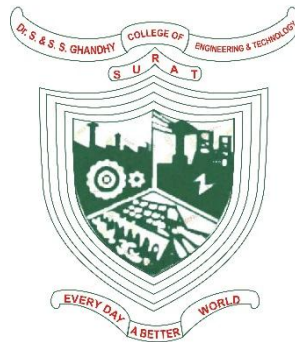
Near-Field Communication

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CERTIFICATE

This is to certify that the report entitled “**NFC - Near Field Communication**” is a bonafied work carried out by **Mr. Krish Ujeniya(216120316028)** under the guidance and supervision of **Mr. Dharmesh Dhangar** for the subject **Seminar (4331605)** of 3rd Semester of Diploma Engineering in **Information Technology** at Dr. S. & S. S. Ghandhy College of Engineering & Technology, Surat. This work has been carried out under my supervision and is to the satisfaction of the department.

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Yours Sincerely,

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ABSTRACT

NFC means Near-field communication. NFC is wireless technology that allows two devices can share any data under 4 cm distance. NFC devices have to type: 1) active device 2) passive device. NFC tags program by many apps. NFC work on radio frequency 13.65 MHz. NFC based on three type operation: 1) Peer to peer mode 2) read/write mode 3) Card emulation. Applications of NFC is Data transfer, Gaming, Identification etc. Compared to Bluetooth NFC is faster data transfer or no need of pairing. NFC in future combine with 5g & IoT makes very faster world.

CHAPTER 1

Introduction

1.1 What is NFC ?

- NFC allows you to share small payloads of data between an NFC tag and an Android-powered device, or between two Android-powered devices.
- NFC is an Extension of Radio frequency identification Technology.
- RFID is used for tracking and identification by sending radio waves.
- NFC is mainly aimed for handheld devices.

1.2 History of NFC

- In 1983 First RFID patent
- In 2004 NFC Forum created by Nokia, Philips and Sony
- In 2006 Nokia launches first NFC phone
- In 2010 Samsung launches first Android NFC phone
- In 2011 Google Wallet launched
- In 2013 Samsung and Visa plans the Mobile payment with NFC

1.3 NFC devices

- NFC devices are two type:-

1.3.1 Active devices :-

- It has a power source and can send and receive data by switching between the normal active (transmitting) state and a passive (receiving) state.
- Example : Mobile



fig 1.1 Mobile

1.3.2 Passive devices :-

- It have no power source and can only connect to active devices.
- Example : NFC tags



fig 1.2 NFC tag

CHAPTER 2

NFC tags

2.1 What is NFC tags

- A NFC tag is a sticker with small microchips that can be read by in range mobile devices. Information is stored in these microchips.
- NFC tags also perform a variety of actions, such as changing handset settings or launching a website.
- You can write data to an NFC tag using an NFC-enabled Mobile phone.
- Most used App :
 1. NFC Tools
 2. NXP TagWriter
 3. Trigger
 4. Simply NFC

2.2 NFC Tools App

Programming With Android Device

You'll need the following to program NFC tags with your Android device:

- An NFC-enabled Android device
- Zapped NFC tag
- An NFC Tag Writing App (NFC Tools is best and you can download NFC Tasks App as an alternative)

If you don't have all of the above, you won't be able to program an NFC tag with your Android device.

The steps include:

1. Install an NFC Tag Writing App on your Android phone. There are many NFC tag-writing apps available on the Google Play Store. We recommend using the NFC Tools app.

2. Open the app and tap on the “Write” button to add a new record for the blank NFC tag.
3. Select the “URL” record type and enter the Zapped vCard link you want to store on the NFC tag. There are three main options: selecting the protocol, filling in the desired URL, and clicking “OK.”
4. Now, place it on your smartphone for the writing part and wait for the encoding process to complete. You can then test the tag by holding your phone near it.
5. Most smartphones are compatible with NFC tags, but you may need to turn on the NFC setting on your phone first.
6. If you encode the tag correctly, you should see a notification on your screen. You can now scan NFC tags if the data has been successfully written to the NFC tag by holding your Android device over the tag.
7. If everything is working, you should see a notification with the text you entered. And that’s it! You have now successfully programmed an NFC tag with your Android device.

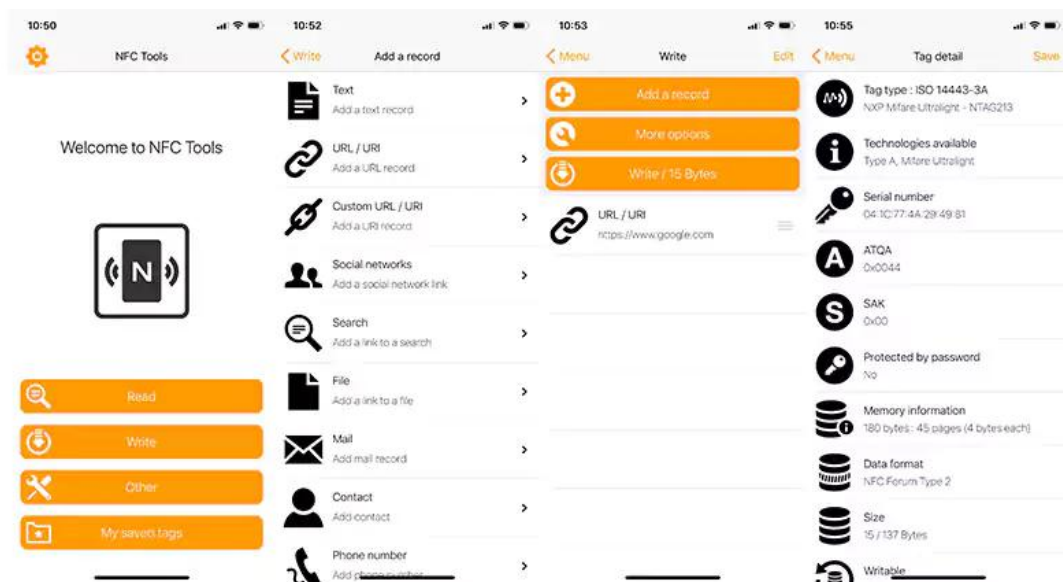


fig 2.1 NFC Tools App

Programming With An iPhone

To program your NFC card on your iPhone, you need to follow the steps:

1. Install “NFC Tools” from the App Store for iPhone. Open NFC Tools app and click “Write.”
2. Click “Add a record.”
3. Select the “URL” option to enter your Zapped vCard URL. Click On “OK.”

4. Hold your iPhone close to the NFC tag you want to encode. A message should appear asking if you're going to encode the tag. Tap "Yes" to continue.
5. Place the prepared NFC tag on the phone's NFC chip location and click "Write."
6. Now, you can read the Tag by approaching it with your phone or by clicking "Read Tag Details" in NFC Tools APP to check all information about the tag written.
7. You can also erase the Tag or lock it for other operations in this interface.

And you're done programming NFC tag successfully. Now you know how to program your own NFC tags with your iPhone. You can use this process to encode any information into an NFC tag, including text, URLs, contact information, and more.

2.3 Program NFC tags with Python

- The `nfcpy` module implements NFC Forum specifications for wireless short-range data exchange with NFC devices and tags.
- It is written in Python and aims to provide an easy-to-use yet powerful framework for applications integrating NFC.
- The source code is licensed under the EUPL and hosted at GitHub. The latest release version can be installed from PyPI with `pip install -U nfcpy`.
- To send a web link to a smartphone:

```
import nfc
import ndef
from threading import Thread
def beam(llc):
    snep_client = nfc.snep.SnepClient(llc)
    snep_client.put_records([ndef.UriRecord('http://nfcpy.org')])
def connected(llc):
    Thread(target=beam, args=(llc,)).start()
    return True
with nfc.ContactlessFrontend('usb') as clf:
    clf.connect(llcp={'on-connect': connected})
```

CHAPTER 3

Working of NFC

- NFC based on inductive coupling. NFC works using magnetic induction between two loop antennas located within each other's near field.
- Operating frequency 13.56 MHz. Data rate 106 kbit/s to 424 kbit/s. NFC use initiator and a target when the initiator actively generates RF field that can power a passive target.

3.1 Peer to peer mode

- Active NFC devices exchange information by switching between passive and active states.



fig 3.1 peer to peer mode

3.2 Read / write mode

- Active NFC device reads or writes information from/on a passive NFC device.



fig 3.2 Read/Write Mode

3.3 Card emulation

- An Active NFC device used for contactless payment.



fig 3.3 Card Emulation

CHAPTER 4

Applications of NFC

4.1 Applications of NFC

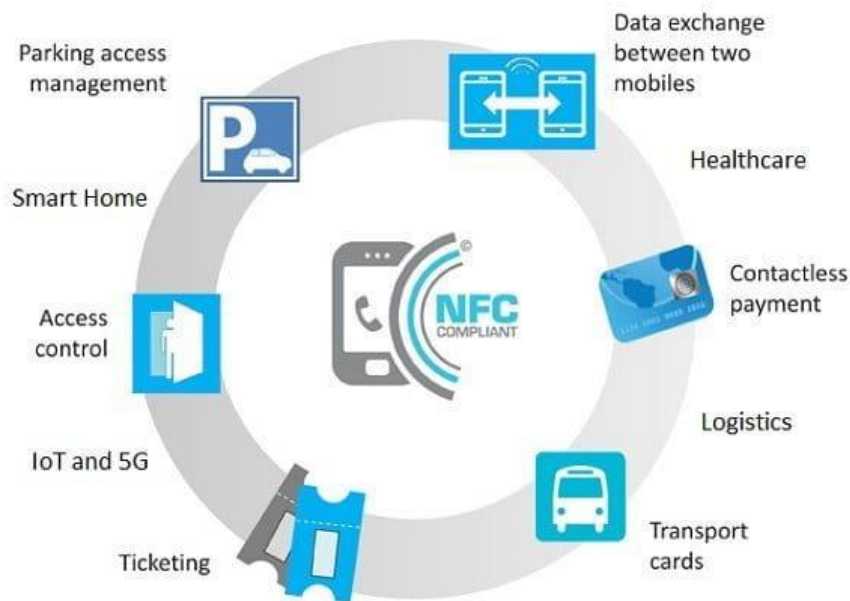


fig 4.1 Applications of NFC

- 1. Data transfer**
- 2. Mobile payments**
- 3. Quick pairing**
- 4. Public transport access**
- 5. Gaming**

4.2 Advantages of NFC

- It is advantageous to perform installments and other transactions utilizing cell phones with the assistance of wallet applications.
- It is utilized for numerous applications in different areas, for example, banking, reservations, booking tickets, recovering, passage/leave passes, and so on.
- It helps the two clients and ventures.

- It gives tied down admittance to understudies and workers in their premises.
- It gives greater security contrast with attractive strip based charge and Mastercards. It additionally utilizes PIN.
- It doesn't need search and pair methodology like Bluetooth and different techniques to build up availability.

4.3 Disadvantages of NFC

- It is extravagant for the organizations to embrace the NFC empowered gadgets.
- It isn't so worthwhile for clients because of complex methods contrast with other simple choices.
- The power utilization is relatively more in NFC empowered gadgets.

Conclusion

NFC tags will play an inevitable role in future smart devices for more integrated functions, smart transportation, aviation industry, shipping, and manufacturing industry for automation of particular tasks. Integrating NFC technology with our modern data communication and transaction process ensures convenience, time-saving, energy efficiency, and most importantly improved security.

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